Department of Mechanical and Production Engineering

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Lecture Notes

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2 Introduction

This collection of notes are the personal notes of me, Noah Hansen, for the course “Theory of Elasticity” (Da: Elasticitetsteori) taught on the 4th semester of the bachelor on the mechanical engineering programme at Aarhus Universitet.

The notes will take base in the lectures by Assistant Professor Matteo Pezzulla and “Introduction to Linear Elasticity” by Phillip L. Gould and Yuan Feng.

2.1 Coordinate Rotation

Given a position vector $\boldsymbol{r}$ to a point P, we can resolve it into components with respect to two different Cartesian systems, x_i and x'_i, having a common origin.

The point P has coordinates $P(x_1, x_2, x_3) = P(x_i)$ in the unprimed system and $P(x'_1, x'_2, x'_3) = P(x'_i)$ in the primed system. The linear transformation between the coordinates is then:

$$
\begin{aligned}
x'_1 &= \alpha_{11}x_1 + \alpha_{12}x_2 + \alpha_{13}x_3 \\
x'_2 &= \alpha_{21}x_2 + \alpha_{22}x_2 + \alpha_{23}x_3 \\
x'_3 &= \alpha_{31}x_1 + \alpha_{32}x_2 + \alpha_{33}x_3
\end{aligned}
$$

or

$$x'_i = \alpha_{ij}x_j.$$

Each of the nine quantities α_{ij} is the cosine of the angle between the ith primed and the jth unprimed axis, i.e.

$$\alpha_{ij} = \cos\left(x'_i, x_j\right) = \frac{\partial x_j}{\partial x'_i} = \boldsymbol{e}'_i \cdot \boldsymbol{e}_j = \cos\left(\boldsymbol{e}'_i, \boldsymbol{e}_j\right).$$

These α_{ij}’s are known as direction cosines.

2.2 Cartesian Tensors

A tensor of order n is a set of 3^n quantities which transform from one coordinate system x_i to another x'_i, by the following law:

Table 1: Tranformation laws for Tensors

n	Order	Transformation law
0	Zero (scalar)	$A'_i = A_i$
1	One (vector)	$A'_i = \alpha_{ij}A_j$
2	Two (dyadic)	$A'_{ij} = \alpha_{ik}, \alpha_{jl}, A_{kl}$
3	Three	$A'_{ijk} = \alpha_{il}\alpha_{jm}\alpha_{kn}A_{lmn}$
4	Four	$A'_{ijkl} = \alpha_{im}\alpha_{jn}\alpha_{kp}\alpha_{lq}A_{mnpq}$

Order zero and order one tensors (scalars and vectors are already familiar quantities, wheres higher-order tensors are not. These tensors are, however, useful to describe physical quantities with a corresponding number of associated directions. Second-order tensors (dyadics) are especially prevalent for elasticity and the corresponding transformation may be carried out in matrix format as:

$$\boldsymbol{A}' = \boldsymbol{R}\boldsymbol{A}\boldsymbol{R}^T.$$

in which

$$\boldsymbol{R} = \begin{bmatrix} \alpha_{11} & \alpha_{12} & \alpha_{13} \\ \alpha_{21} & \alpha_{22} & \alpha_{23} \\ \alpha_{31} & \alpha_{32} & \alpha_{33} \end{bmatrix} \qquad \text{and} \qquad \boldsymbol{A}' = \begin{bmatrix} A'_{11} & A'_{12} & A'_{13} \\ A'_{21} & A'_{22} & A'_{23} \\ A'_{31} & A'_{32} & A'_{33} \end{bmatrix}.$$

2.2.1 Algebra of Cartesian Tensors

Arithmetic and algebra for tensors is similar to matrix operations when it comes to addition, subtraction, equality, and scalar multiplication. Multiplication of two tensors of order n and m produces a now tensor of order $n+m$, e.g.

$$A_i B_{jk} = C_{ijk}.$$

2.3 Operational Tensors

Additional tensor operations are made possible by the Kronecker delta δ_{ij}, defined as,

$$\delta_{ij} \equiv \begin{cases} 1 & i = j \\ 0 & i \neq j. \end{cases}$$

and the permutation tensor ϵ_{ijk} defined such that:

$$\begin{aligned} \epsilon_{ijk} &= \frac{1}{2}(i-j)(j-k)(k-i) \\ &= \begin{cases} 0 & \text{If any two of } i, j, k \text{ are equal} \\ 1 & \text{For an even permutation (forward on the number line } 1,2,3,1,2,3,\ldots) \\ -1 & \text{For an odd permutation (backward on the number line } 3,2,1,3,2,1,\ldots). \end{cases} \end{aligned}$$

The Kronecker delta δ_{ij} changes the subscripts in a tensor by multiplication:

$$\begin{aligned} \delta_{ij} A_i &= \delta_{1j} A_1 + \delta_{2j} A_2 + \delta_{3j} A_3 = A_j \\ \delta_{ij} C_{ijk} &= \delta_{11} C_{11k} + \delta_{22} C_{22k} + \delta_{33} C_{33k} = C_{iik} = A_K. \end{aligned}$$

It is also useful in vector algebra and for coordinate transformations. E.g. starting with the dot product of two unit vectors:

$$\boldsymbol{e}_i \cdot \boldsymbol{e}_j = \delta_{ij} \tag{1}$$

and seeking the component of vector $\boldsymbol{A}$ in the j-direction A_j we do:

$$\begin{aligned} \boldsymbol{e}_j \cdot \boldsymbol{A} &= \left(\boldsymbol{e}_j \cdot \boldsymbol{e}_i\right) A_i \\ &= \delta_{ji} A_i \\ &= A_j. \end{aligned}$$

The permutation tensor ϵ_{ijk} is useful for vector cross-product operations:

$$\epsilon_{ijk} A_j B_k \boldsymbol{e}_i = \boldsymbol{A} \times \boldsymbol{B}.$$